Daily Drilling Report
SPD19B-03
9/4/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
39.0	12 1/4" & 8 1/2"	184.00	72.83	31.70	92.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
9 5/8" Shoe, 2,720.80mKB	220,877.90	24,399.10	9,311,204.92		34.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
2,725.00	2,725.00	0.00			

Operations at Report Time
Drilling @ 2725m.

Operations Summary
R/D CMT head. B/O HGR RT. R/D CSG handling tools. Wash inside WH. Set seal pack off. L/D old 12 1/4" BHA. Perform BOP test. Make 10 stands 5 7/8" DP & 1 stand HWDP. RIH new 8 1/2" directional BHA to 1900m.

Operations next 24hrs
Cont' RIH directional BHA#10 and tag plugs @ 2696m. D/O shoe track and rathole to 2725m. Drill new formation to 2728m. Perfotm FIT. Drilling 8 1/2" hole section.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Sajad Tamaddon	Senior Drilling Sup.
Vahab Kari / Behnam Jahangiri	Night Drilling Sup.
Moslem Yarahmadi	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	8/23/2014	10/13/2014
BOP Test	8/20/2014	9/11/2014
Chemical Spill Drill	8/17/2014	10/17/2014
Fire Drill	8/23/2014	10/13/2014
H2S Drill	8/31/2014	10/1/2014
Heli Crash Drill	7/15/2014	9/14/2014
Medical Drill	7/26/2014	9/25/2014
MOB Drill	9/1/2014	11/1/2014
Pit Drill	8/25/2014	10/10/2014
Trip Drill	9/4/2014	10/25/2014

STOP Cards

Number of Reports: 30

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	00:30	0.50	12 1/4"	H	01	P	B/O surface lines and R/D Mehran CMT Head & X/O. Meanwhile wash inside WH through side outlet valve by CMT unit.
00:30	01:00	0.50	12 1/4"	G	01	P	B/O & L/D landing jt, CSG hanger R/T + 2 x Pup Jts as one piece.
01:00	02:30	1.50	12 1/4"	G	01	P	R/D Al-Ghaith equipment. R/U 5 7/8" pipe handling equipment.
02:30	03:30	1.00	12 1/4"	C	01	P	P/U & M/U AKER 13 5/8" washing tool to one std 5 7/8" DP, RIH and wash inside BOP and Unihead and top of 10 3/4" CSG HGR w/ 600 gpm , 5 rpm, 60 psi pressure. Meanwhile wash inside WH through side outlet valves by CMT unit. POOH and L/D washing tool.
03:30	05:00	1.50	12 1/4"	C	05	P	P/U & M/U 10 3/4" Aker Pack-off Assy + R/T+ one std of 5 7/8" DP. RIH to 18.8m, L/U CMT unit to Choke line, flush surface lines and pressure test same to 500/5000 psi for 5/10 min, OK. Close Upper Ram. Set Pack-off Assy by apply 2000 psi pressure in annulus. Increase pressure to 5000 psi for energizing, OK. Pull with 10 Kips over pull and release RT. POOH to surface, B/O & L/D R/T. Perform pressure test elastomeric seal area from Uni-Head port by hand pump to 5000 psi/10 min, OK.
05:00	09:00	4.00	12 1/4"	D	04	P	L/D old 12 1/4" motor BHA consisting of: 9 5/8" Motor + 8" Pony NMDC + 11 7/8" String Stabilizer + 1 jt 8" NMDC + float sub +8 1/4" DC + X/OVER and 12 1/4" Rotary BHA consisting of 3 joints 8 1/4" DC + 12 1/8" String stabilizer
09:00	12:30	3.50	8 1/2"	C	12	P	M/U test assembly consisting of 13 5/8" Aker Spacer + 13 5/8" Aker Test Plug + XOS (4 1/2" IF into 5 7/8" XT57) + 2 joints 5 7/8" DP + Side entry sub + 2 x TIW +10 ft Pup joint, RIH to 18.8m. Connect CMT unit to CMT manifold line. L/U to CMT unit. Flush lines, pressure test to 500/5000 psi for 5/10 min, ok. Perform 18 3/4" BOP test (Low test: 500 psi for 5 min , High test: 5000 psi for 10 min) as per below, all OK. 1) Lower pipe ram/Lower TIW 2) VBR/Upper TIW/HCR choke 3) Upper pipe ram/Manual Choke/ HCR kill/ Upper IBOP 4) Annular/Manual Kill/Lower IBOP 5) Blind/Shear ram RIH and retrieve the test plug. B/out and L/down same.
12:30	13:00	0.50	8 1/2"	D	01	P	P/U and M/U Aker wear bushing running tool. RIH and set short wear bushing inside unihead.
13:00	16:00	3.00	8 1/2"	D	01	P	P/U and M/U 30 jts 5 7/8" DP & 4 jts 5 7/8" HWDP from the main deck, RIH and flush inside. POOH & Rack same in derrick.

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Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
16:00	18:00	2.00	8 1/2"	D	04	P	P/U & M/U 8 1/2" directional BHA#10 consisting of below items and RIH to 25m. Align Motor with MWD. 1) 8 1/2" Varel PDC bit (Type: VM713GUX, IADC: M332, SN: 6006867, w/ 4 x 14/32" nozzles) 2) 7" Motor (Bent housing: 1.15°, Sleeve Stabilizer: 8 3/8" , Rev./gal: 0.29 , Lobes: 6/7 , Stage: 5) 3) 6 13/16" PNMDC 4) 8" COR Stabilizer (SN: COR-2491-AD) 5) 6 3/4" NMDC 6) 6 3/4" MWD TMS
18:00	18:30	0.50	8 1/2"	D	12	P	Connect TDS & perform MWD & Motor shallow hole test w/ 500 gpm/ 780 psi, ok.

18:30	19:30	1.00	8 1/2"	D	04	P	P/U the rest of BHA#10 consisting of below items and RIH to 256m.
							
							07) 6 3/4" NMDC
							08) 6 3/4 F. Sub
							09) XOS (4 1/2" IF pin x XT57 box)
							10) 20 Jts. 5 7/8" HWDP
							11) 7" drilling jar
							12) 3 Jts. 5 7/8" HWDP
							
							>>> Weight below jar (in mud): 25.7 kips
							>>> BHA weight (in mud): 29.6 kips

19:30	00:00	4.50	8 1/2"	D	04	P	RIH directional drilling BHA#10 on 5 7/8" DP f/ 256m to 1900m. *** Fill up string with old barite mud every 15 STDs.
00:00	00:00		8 1/2"				Sep 05 , 2014 00:00-03:30 Cont' RIH dir. drilling BHA#10 on 5 7/8" DP f/ 1900m to 2664m. Connect TDS, W/D w/ 300gpm / 1000psi, tag top cement plug @ 2696m. 03:30-06:00 D/O CMT plugs + Float collar (2696.5m) + cement inside shoe track + Float shoe (2720.8m) and clear rathole to 2725m w/ 450 gpm / 1850 psi / 2 - 10 WOB / 30 rpm / 14 - 18 Kft-lbf. *** Pump 30 bbls Hi-Vis pill (from old barite mud) after D/O float collar. *** Check shoe with and without rotation, OK. *** Pump 50 bbl Hi-Vis pill + 100 bbl SW + 650 bbl new CaCo3/KCL/ Polymer 10.5 ppg mud and displace hole while D/O cement.

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
2,725.20	CaCo3/KCL/ Polymer	10.50	46	20.0	15.000	5.5
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
9.5	0.300	1.600	1.200			
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
3	4	0.0	86.0	14.0	33,000.000	1
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
5.000	6.000	7.000	0/100		760.0	2624.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
6074.0	20.0	0.0			7.244419	0.459

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
0.30	6.	245	7.0	996.0	36.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
EP-HUA	Chopper	9/4/2014 11:39	9/4/2014 11:46	Pax in:01, Pax out:02
Borkan	Supply Boat	9/4/2014 13:00	9/4/2014 13:15	Transfrd 03 pxs for Sahar 1, C/Off & Dep for Sahar 1
CB FADAK-20	Supply Boat	9/4/2014 19:10	9/4/2014 20:10	Arrived form Kish, Embarked 08 PAX (07 PAX for Kish + 01 PAX for Aban-8), departed for Sahar-1

Mud Additives

Description	Consumed	Cum Rec	On Loc
Caustic Soda 2014	14.0	340.0	156.0
Glydril	16.0	100.0	40.0
MI-Cide	2.0	47.0	35.0
Starch	120.0	1,200.0	97.0
Pac-LV	70.0	481.0	65.0
Soda Ash 25 Kg	20.0	550.0	177.0
Zinc Carbonate	40.0	119.0	79.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Barite BB	0.0	0.0	0.0
Potable Water	23.0	12.0	74.0
Drill Water	13.0		330.0
Cement G	0.0	0.0	92.0

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Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Diesel Fuel	7.0		160.0
Barite Bulk	0.0	0.0	129.0
Bentonite Bulk	0.0	0.0	8.0
Bentonite BB	0.0	0.0	0.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
10	8 1/2	Varel	VM713GUX	6006867	PDC	M332	14/14/14/14
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
0.60	0.00						-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
3 jts 5 7/8" HWDP	5 7/8	4.000	27.89	255.95
7" Drilling Jar (COR-3021AD)	7	2.938	5.80	228.06
20 jts 5 7/8" HWDP	5 7/8	4.000	186.35	222.26
X/O	6 1/2	2.813	0.85	35.91
6 3/4" Float sub	6 3/4		0.81	35.06
6 3/4" NMDC	6 3/4	3.250	9.48	34.25
6 3/4" TM - Sub	6 3/4		1.74	24.77
6 3/4" NMDC	6 3/4	3.250	9.48	23.03
8" St. Stabilizer (COR-2491AD)	8	3.000	1.84	13.55
Pony NMDC	6 13/16	3.500	3.09	11.71
6 3/4" Motor (AKO: 1.15° , Sleeve: 8 3/8")	6 3/4		8.34	8.62
8 1/2" PDC bit	8 1/2		0.28	0.28

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
400.0	8.0	400.00		11.48	1,650.4
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
8/21/2014	13 3/8" Shoe, 1,487.00mKB	Formation Integrity	1,495.00	1,427.35	1,200.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,200.0	7.7	1,200.00		14.03	3,413.7

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
S4		2,594.00	2,457.78
S3		2,622.00	2,483.90
S2		2,632.50	2,493.72
S1		2,675.70	2,534.06
Upper Kangan		2,703.00	

General Remarks

Comment

>> Requirements: (MI MR # 14 & 15)

>>> Sent:

** Forte to Kish: (Starch Samples) + (12 x Casing, 9 5/8", VAM TOP/Blue) + (28 x CASING, 9 5/8" 47 PPF, TMK PF) + (19 x CASING, 10 3/4", 55.5 PPF, L 80, API 5CT, TYPE 1, TENARIS) + (5 x CASING, 10.3/4 ", VAM TOP) + (2 x XO, 9 5/8" 47 PPF VAM TOP PIN X 9 5/8" 47 PPF TMK PF BOX) + (2 x XO, 10 3/4", 60.7 PPF, VAM TOP BOX X 9 5/8" 47 PPF TMK PF PIN) + (1 x XO, 10 3/4" 55.5 PPF TENARIS PIN X 10 3/4" 55.5 PPF VAM TOP BOX) + (2 x XO 10 3/4" 55.5 PPF VAM TOP PIN x 10 3/4" 55.5 PPF TENARIS BOX) + (28 x CENTRALIZER, Botil , BOW, 9 5/8"X 12.1/4) + (69 x STOP COLLAR, 9 5/8", Screw Type) + (1 x CENTRALIZER, Botil , RIGID TYPE, 9 5/8"X 13 3/8") + (4 x PUP JOINT,9 5/8",VAM TOP ,47 PPF,L 80,2.5 & 4 MT LONG) +(5 x PUP JOINT,10 3/4", VAM TOP,55.5 PPF,2, 5, 3, 2.5 & 7 MT LONG) + (2 x 9 5/8" F. Shoe) + (2 x 9 5/8" F. Collar) + (10 x Casing Couplings) + (1 x CEMENT PLUG,TOP PLUG,10 3/4"X 9 5/8",47 & 1 x CEMENT PLUG, BTM PLUG,10 3/4"X 9 5/8",47) + (2 x long wear bushing) + (2 x pack off ass. 13 5/8" x 10 3/4") + (2 x Pack off RT) + (2 x Packoff Ret/T) + (2 x SETTING TOOL emergency,13 5/8") + (1 x R&R TOOL FOR EMERGENCY PACKOFF 13 5/8" x 10 3/4) + (6 x Pack off, METAL SEAL, 13 5/8 X 10 3/4 CSG) + (2 x SEAL, FOR EMERGENCY PACKOFF, 13 3/8" X 10 3/4,METAL SEAL) + (2 x EMERGENCY PACKOFF ASSEMBLY, 13 5/8" NOMINAL X 10 3/4") + (2 x EMERGENCY SLIP SPIDER ASSEMBLY, 13 5/8" NOMINAL X 10 3/4) + (1 x Bit, PDC, 12 ¼") + (1 x CROSSOVER, 9 5/8" 47 PPF VAM TOP PIN X 10 3/4" 55.5 PPF, TENARIS BLUE BOX) + (3 x 9 5/8" casing circulating swage) +(1 x 10 3/4" circulating swage) + (2 x casing drift 9 5/8") + (2 x casing drift 10 3/4") + (casing protectors) + (1x 12 1/8" Stab) + (1 x 11 7/8" stab) + (7" jar) + (Garbage) + (1 x CSG Hanger 10 3/4") + (2 x Hanger RT) + (2 x CMT Head) + (2 x Mehran Swages & valves) ....

>>> Received:

** Nill

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General Remarks

Comment

..... + (HPOGC 2 x 9 5/8" Mud Motor, 4 x 8" NMDC, 8 1/2" Fsub, 2 x 12 1/8", 2 x 12 1/16", 1 x 12" sleeve stab., 1 x PNMDC, 2 x 8" restrictor sub, 1 x 8" Fsub, 2 x float valve 5F/6R, 2 x Transmitter sub, 1 x APC protector, 1 x box batteries & 1 x wooden box containing transmitter & clamp bolt & EOR)